ASSIGNMENT No: 05 (BI)
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Title: Machine Learning for Genomic Data
Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import
accuracy_score,precision_score,recall_score,f1_score,confusion_matrix,classification_report

print("MACHINE LEARNING FOR GENOMIC DATA")
print("-"*60)

np.random.seed(42)

data={
   "Gene_1":[10,12,11,13,9,80,85,82,90,88,15,14,13,12,11,75,78,80,82,85],
   "Gene_2":[80,85,82,88,90,15,12,18,14,16,78,80,82,85,88,10,15,12,18,14],
   "Gene_3":[20,22,19,21,23,70,72,68,75,73,25,24,22,23,21,65,68,70,72,74],
   "Gene_4":[30,32,28,31,29,60,65,62,68,64,35,34,32,30,33,58,62,65,67,63],
   "Gene_5":[15,18,16,17,14,90,88,92,85,87,20,19,17,18,16,82,85,88,90,86],
   "Class":[0,0,0,0,0,1,1,1,1,1,0,0,0,0,0,1,1,1,1,1]
}

df=pd.DataFrame(data)

print("\nGenomic Dataset:")
print(df)

print("\nDataset Shape:")
print(df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

X=df.drop("Class",axis=1)
y=df["Class"]

print("\nFeatures:")
print(X.columns.tolist())
```

```
print("\nTarget Classes:")
print(y.unique())

X_train,X_test,y_train,y_test=train_test_split(
   X,y,test_size=0.30,random_state=42,stratify=y
)

print("\nTraining Samples:",len(X_train))
print("Testing Samples:",len(X_test))

scaler=StandardScaler()
X_train_scaled=scaler.fit_transform(X_train)
X_test_scaled=scaler.transform(X_test)

rf_model=RandomForestClassifier(n_estimators=100,random_state=42)
rf_model.fit(X_train,y_train)
rf_predictions=rf_model.predict(X_test)

svm_model=SVC(kernel="linear",random_state=42)
svm_model.fit(X_train_scaled,y_train)
svm_predictions=svm_model.predict(X_test_scaled)

print("\n\nRANDOM FOREST RESULTS")
print("-"*60)

rf_accuracy=accuracy_score(y_test,rf_predictions)
rf_precision=precision_score(y_test,rf_predictions,zero_division=0)
rf_recall=recall_score(y_test,rf_predictions,zero_division=0)
rf_f1=f1_score(y_test,rf_predictions,zero_division=0)

print("Accuracy :",rf_accuracy)
print("Precision:",rf_precision)
print("Recall   :",rf_recall)
print("F1-Score :",rf_f1)

print("\nClassification Report:")
print(classification_report(y_test,rf_predictions,zero_division=0))

print("\n\nSVM RESULTS")
print("-"*60)

svm_accuracy=accuracy_score(y_test,svm_predictions)
svm_precision=precision_score(y_test,svm_predictions,zero_division=0)
svm_recall=recall_score(y_test,svm_predictions,zero_division=0)
svm_f1=f1_score(y_test,svm_predictions,zero_division=0)
```

```
print("Accuracy :",svm_accuracy)
print("Precision:",svm_precision)
print("Recall   :",svm_recall)
print("F1-Score :",svm_f1)

print("\nClassification Report:")
print(classification_report(y_test,svm_predictions,zero_division=0))

rf_cm=confusion_matrix(y_test,rf_predictions)
svm_cm=confusion_matrix(y_test,svm_predictions)

plt.figure(figsize=(6,5))
sns.heatmap(rf_cm,annot=True,fmt="d")
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6,5))
sns.heatmap(svm_cm,annot=True,fmt="d")
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()

importance=rf_model.feature_importances_

feature_importance=pd.DataFrame({
   "Feature":X.columns,
   "Importance":importance
})

feature_importance=feature_importance.sort_values(
   by="Importance",ascending=False
)

print("\n\nFEATURE IMPORTANCE")
print("-"*60)
print(feature_importance)

plt.figure(figsize=(8,5))
plt.bar(feature_importance["Feature"],feature_importance["Importance"])
plt.xlabel("Genomic Features")
plt.ylabel("Importance")
plt.title("Random Forest Feature Importance")
```

```
plt.tight_layout()
plt.show()

comparison=pd.DataFrame({
   "Model":["Random Forest","SVM"],
   "Accuracy":[rf_accuracy,svm_accuracy],
   "Precision":[rf_precision,svm_precision],
   "Recall":[rf_recall,svm_recall],
   "F1-Score":[rf_f1,svm_f1]
})

print("\n\nMODEL COMPARISON")
print("-"*60)
print(comparison.round(4))

print("\n\nRESULT")
print("-"*60)

if rf_f1>svm_f1:
   print("Random Forest performed better based on F1-Score.")
elif svm_f1>rf_f1:
   print("SVM performed better based on F1-Score.")
else:
   print("Both models performed equally based on F1-Score.")

print("\nMachine learning classification completed successfully.")
print("Aditya Rade   BEAD23167")
```

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OUTPUT:

```
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\AppDa
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Pracical-5-BI.py"
MACHINE LEARNING FOR GENOMIC DATA
------------------------------------------------------------

Genomic Dataset:
    Gene_1  Gene_2  Gene_3  Gene_4  Gene_5  Class
0       10      80      20      30      15      0
1       12      85      22      32      18      0
2       11      82      19      28      16      0
3       13      88      21      31      17      0
4        9      90      23      29      14      0
5       80      15      70      60      90      1
6       85      12      72      65      88      1
7       82      18      68      62      92      1
8       90      14      75      68      85      1
9       88      16      73      64      87      1
10      15      78      25      35      20      0
11      14      80      24      34      19      0
12      13      82      22      32      17      0
13      12      85      23      30      18      0
14      11      88      21      33      16      0
15      75      10      65      58      82      1
16      78      15      68      62      85      1
17      80      12      70      65      88      1
18      82      18      72      67      90      1
19      85      14      74      63      86      1

Dataset Shape:
(20, 6)
```

```
Missing Values:
Gene_1    0
Gene_2    0
Gene_3    0
Gene_4    0
Gene_5    0
Class     0
dtype: int64

Features:
['Gene_1', 'Gene_2', 'Gene_3', 'Gene_4', 'Gene_5']

Target Classes:
[0 1]

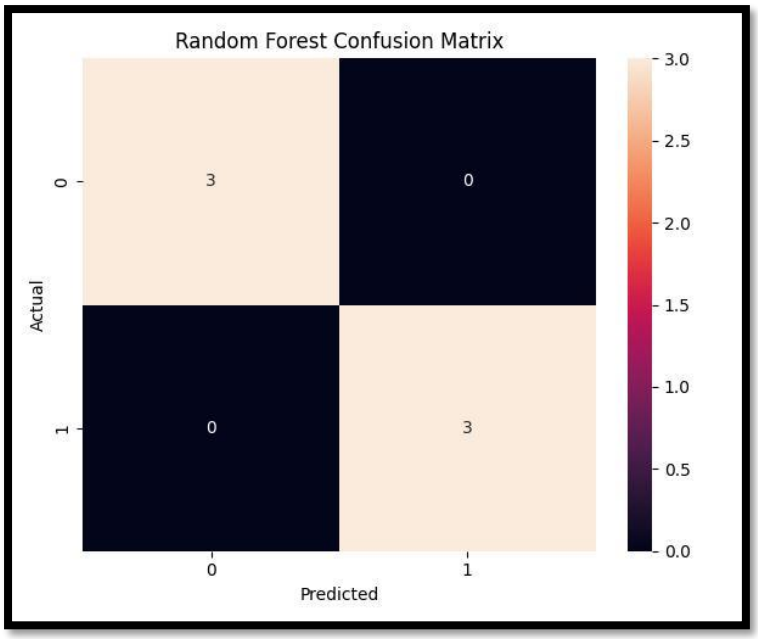
Training Samples: 14
Testing Samples: 6
```

```
RANDOM FOREST RESULTS
------------------------------------------------------------
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1-Score : 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         3

    accuracy                           1.00         6
   macro avg       1.00      1.00      1.00         6
weighted avg       1.00      1.00      1.00         6
```




```
SVM RESULTS
--------------------------------------------------------------------
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1-Score : 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         3

    accuracy                           1.00         6
   macro avg       1.00      1.00      1.00         6
weighted avg       1.00      1.00      1.00         6
```